Supplementary code: generating a logistic-regression sample with a known population R-squared

Purpose

This file illustrates, for a single case – a logistic regression whose population R-squared is exactly known in advance – the calibrate-then-sample method used throughout the simulations reported in the paper. The code below is a stripped-down extract of the logistic branch of `sample_by_r2()` from the `Rsimcity` package (https://github.com/mcfanda/RsimCity): input validation, caching, and the branches for the other model families (gaussian, multinomial, ordinal) and for targeting a single predictor's eta-squared rather than the whole model's R-squared have all been removed, so that only the logic essential to this one case remains. All of those variations, their defaults, and their full implementations can be found in the package itself.

The method in three steps

1. Build a numerical-integration grid over the joint distribution of the `k` predictors, using Gauss-Hermite quadrature.
2. Find, by root-finding, the coefficient scale that makes the *population* R-squared – computed exactly on that grid – equal the requested target.
3. Only then draw an actual finite sample of size `n` from the calibrated population.

Code

```
sample_logistic_r2 <- function(n, R2, k = 1, beta = rep(1, k),
                               intercept = 0, x_mean = rep(0, k), x_cov = diag(k),
                               order = 10, seed = NULL) {

  ## Gauss-Hermite quadrature grid over the k predictors
  ## using statmod::gauss.quad.prob(). It returns nodes/weights for a standard normal density,
  ## with weights already summing to 1, so that for any function f, E[f(X)] = sum(w * f(Xgh)).
  ## The grid is then transformed through the Cholesky factor of x_cov and shifted by x_mean, giving qu
  ## for the actual (correlated, possibly non-centered) predictor distribution.
  gh <- statmod::gauss.quad.prob(order, dist = "normal")
  g  <- as.matrix(expand.grid(rep(list(seq_len(order)), k)))
  Z  <- matrix(gh$nodes[g], ncol = k)
  w  <- apply(matrix(gh$weights[g], ncol = k), 1, prod)
  Xgh <- sweep(Z %*% chol(x_cov), 2, x_mean, "+")

  ## Population deviance R-squared, as a function of a coefficient scale `a`
  ## beta = a * beta direction; the model's probability at every grid node gives the
  ## deviance-based R-squared exactly (no simulation is involved in this computation).
  r2_of_a <- function(a) {
    eta  <- intercept + drop(Xgh %*% (a * beta))
    prob <- stats::plogis(eta)

    p    <- cbind(1 - prob, prob)
    p    <- pmax(p, .Machine$double.eps)
    p0   <- colSums(p * w)                         # population marginal probabilities
```

```r
    D1   <- -2 * sum(w * rowSums(p * log(p)))      # model deviance
    D0   <- -2 * sum(p0 * log(p0))                 # null-model deviance
    1 - D1 / D0
  }

  ## ---- 3. Root-find the coefficient scale that produces the target R2 ----------------
  upper <- 1
  while (r2_of_a(upper) < R2 && upper < 1e6) upper <- upper * 2
  a <- stats::uniroot(function(a) r2_of_a(a) - R2, c(0, upper))$root
  beta_final <- a * beta

  ## ---- 4. Draw the actual finite sample from the calibrated population ---------------
  if (!is.null(seed)) set.seed(seed)
  X <- sweep(matrix(stats::rnorm(n * k), n, k) %*% chol(x_cov), 2, x_mean, "+")
  colnames(X) <- paste0("x", seq_len(k))
  eta <- intercept + drop(X %*% beta_final)
  y   <- stats::rbinom(n, 1, stats::plogis(eta))

  data.frame(y = y, X)
}
```

Example: a single predictor, R-squared = .10

```r
dat <- sample_logistic_r2(n = 2000, R2 = .10, seed = 1)
mod <- glm(y ~ x1, data = dat, family = binomial())
1 - mod$deviance / mod$null.deviance   # sample deviance R2, close to the .10 target
```

```
## [1] 0.09601619
```

Example: three correlated predictors, R-squared = .20

```r
Sigma <- matrix(c(1, .3, .2,
                  .3, 1, .4,
                  .2, .4, 1), nrow = 3)

dat3 <- sample_logistic_r2(n = 5000, R2 = .20, k = 3, x_cov = Sigma, seed = 1)
mod3 <- glm(y ~ x1 + x2 + x3, data = dat3, family = binomial())
1 - mod3$deviance / mod3$null.deviance   # close to the .20 target
```

```
## [1] 0.1916439
```

Simulation results tables

The figures in the main text plot the results of Study 1 (recovery of the whole-model R^2 and of a single focal predictor's eta-squared) and Study 2 (simulation-based required sample size). The tables below report, in full, the same numbers shown in those figures.

Study 1a: accuracy of R^2 (data underlying Figure 1)

Table 1: Estimated R2 and adjusted R2 vs. population R2, by model and sample size.

Model	N	Population R2	R2	Adjusted R2
Gaussian	25	0.025	0.147	0.025
Gaussian	25	0.050	0.166	0.046
Gaussian	25	0.075	0.187	0.071
Gaussian	25	0.100	0.209	0.096
Gaussian	25	0.125	0.229	0.119
Gaussian	25	0.150	0.249	0.141
Gaussian	25	0.175	0.270	0.166
Gaussian	25	0.200	0.293	0.191
Gaussian	25	0.225	0.310	0.211
Gaussian	25	0.250	0.330	0.234
Gaussian	25	0.275	0.351	0.259
Gaussian	25	0.300	0.373	0.283
Gaussian	25	0.325	0.394	0.307
Gaussian	25	0.350	0.413	0.329
Gaussian	25	0.375	0.435	0.354
Gaussian	25	0.400	0.457	0.379
Gaussian	50	0.025	0.083	0.023
Gaussian	50	0.050	0.107	0.049
Gaussian	50	0.075	0.129	0.072
Gaussian	50	0.100	0.152	0.096
Gaussian	50	0.125	0.176	0.122
Gaussian	50	0.150	0.198	0.146
Gaussian	50	0.175	0.220	0.169
Gaussian	50	0.200	0.243	0.193
Gaussian	50	0.225	0.263	0.215
Gaussian	50	0.250	0.289	0.242
Gaussian	50	0.275	0.313	0.269
Gaussian	50	0.300	0.335	0.291
Gaussian	50	0.325	0.356	0.314
Gaussian	50	0.350	0.380	0.340
Gaussian	50	0.375	0.406	0.367
Gaussian	50	0.400	0.432	0.394
Gaussian	75	0.025	0.063	0.024
Gaussian	75	0.050	0.086	0.048
Gaussian	75	0.075	0.111	0.074
Gaussian	75	0.100	0.135	0.098
Gaussian	75	0.125	0.158	0.123
Gaussian	75	0.150	0.179	0.144
Gaussian	75	0.175	0.205	0.171
Gaussian	75	0.200	0.228	0.196
Gaussian	75	0.225	0.251	0.219
Gaussian	75	0.250	0.276	0.245
Gaussian	75	0.275	0.301	0.272
Gaussian	75	0.300	0.325	0.296
Gaussian	75	0.325	0.344	0.316
Gaussian	75	0.350	0.372	0.346
Gaussian	75	0.375	0.393	0.368
Gaussian	75	0.400	0.418	0.393
Gaussian	100	0.025	0.055	0.025
Gaussian	100	0.050	0.079	0.050
Gaussian	100	0.075	0.102	0.073
Gaussian	100	0.100	0.127	0.100
Gaussian	100	0.125	0.149	0.123
Gaussian	100	0.150	0.175	0.149
Gaussian	100	0.175	0.198	0.173

Table 1: Estimated R2 and adjusted R2 vs. population R2, by model and sample size. *(continued)*

Model	N	Population R2	R2	Adjusted R2
Gaussian	100	0.200	0.221	0.197
Gaussian	100	0.225	0.244	0.220
Gaussian	100	0.250	0.268	0.245
Gaussian	100	0.275	0.293	0.271
Gaussian	100	0.300	0.317	0.296
Gaussian	100	0.325	0.341	0.320
Gaussian	100	0.350	0.366	0.346
Gaussian	100	0.375	0.387	0.368
Gaussian	100	0.400	0.412	0.393
Logistic	25	0.025	0.127	0.038
Logistic	25	0.050	0.156	0.066
Logistic	25	0.075	0.180	0.091
Logistic	25	0.100	0.206	0.116
Logistic	25	0.125	0.233	0.144
Logistic	25	0.150	0.260	0.170
Logistic	25	0.175	0.281	0.192
Logistic	25	0.200	0.305	0.215
Logistic	25	0.225	0.326	0.237
Logistic	25	0.250	0.348	0.258
Logistic	25	0.275	0.375	0.286
Logistic	25	0.300	0.395	0.306
Logistic	25	0.325	0.417	0.328
Logistic	25	0.350	0.440	0.350
Logistic	25	0.375	0.457	0.368
Logistic	25	0.400	0.475	0.386
Logistic	50	0.025	0.072	0.028
Logistic	50	0.050	0.096	0.052
Logistic	50	0.075	0.123	0.079
Logistic	50	0.100	0.148	0.104
Logistic	50	0.125	0.174	0.130
Logistic	50	0.150	0.201	0.157
Logistic	50	0.175	0.225	0.181
Logistic	50	0.200	0.251	0.207
Logistic	50	0.225	0.275	0.231
Logistic	50	0.250	0.299	0.256
Logistic	50	0.275	0.326	0.282
Logistic	50	0.300	0.355	0.311
Logistic	50	0.325	0.376	0.332
Logistic	50	0.350	0.401	0.357
Logistic	50	0.375	0.428	0.384
Logistic	50	0.400	0.450	0.406
Logistic	75	0.025	0.056	0.026
Logistic	75	0.050	0.081	0.052
Logistic	75	0.075	0.106	0.077
Logistic	75	0.100	0.132	0.103
Logistic	75	0.125	0.156	0.127
Logistic	75	0.150	0.181	0.152
Logistic	75	0.175	0.207	0.178
Logistic	75	0.200	0.233	0.204
Logistic	75	0.225	0.258	0.229
Logistic	75	0.250	0.283	0.254
Logistic	75	0.275	0.307	0.278
Logistic	75	0.300	0.334	0.305
Logistic	75	0.325	0.361	0.332
Logistic	75	0.350	0.384	0.354

Table 1: Estimated R2 and adjusted R2 vs. population R2, by model and sample size. *(continued)*

Model	N	Population R2	R2	Adjusted R2
Logistic	75	0.375	0.406	0.377
Logistic	75	0.400	0.431	0.402
Logistic	100	0.025	0.047	0.026
Logistic	100	0.050	0.073	0.051
Logistic	100	0.075	0.099	0.077
Logistic	100	0.100	0.123	0.102
Logistic	100	0.125	0.148	0.126
Logistic	100	0.150	0.173	0.151
Logistic	100	0.175	0.199	0.177
Logistic	100	0.200	0.224	0.202
Logistic	100	0.225	0.249	0.227
Logistic	100	0.250	0.275	0.253
Logistic	100	0.275	0.299	0.277
Logistic	100	0.300	0.324	0.302
Logistic	100	0.325	0.348	0.327
Logistic	100	0.350	0.375	0.354
Logistic	100	0.375	0.399	0.377
Logistic	100	0.400	0.423	0.401
Multinomial	25	0.025	0.161	0.047
Multinomial	25	0.050	0.187	0.074
Multinomial	25	0.075	0.212	0.098
Multinomial	25	0.100	0.234	0.120
Multinomial	25	0.125	0.262	0.148
Multinomial	25	0.150	0.281	0.167
Multinomial	25	0.175	0.305	0.191
Multinomial	25	0.200	0.326	0.212
Multinomial	25	0.225	0.345	0.230
Multinomial	25	0.250	0.366	0.251
Multinomial	25	0.275	0.384	0.269
Multinomial	25	0.300	0.402	0.287
Multinomial	25	0.325	0.414	0.299
Multinomial	25	0.350	0.429	0.314
Multinomial	25	0.375	0.442	0.327
Multinomial	25	0.400	0.452	0.337
Multinomial	50	0.025	0.086	0.031
Multinomial	50	0.050	0.112	0.057
Multinomial	50	0.075	0.138	0.083
Multinomial	50	0.100	0.163	0.107
Multinomial	50	0.125	0.190	0.134
Multinomial	50	0.150	0.214	0.158
Multinomial	50	0.175	0.241	0.185
Multinomial	50	0.200	0.264	0.208
Multinomial	50	0.225	0.291	0.235
Multinomial	50	0.250	0.316	0.259
Multinomial	50	0.275	0.338	0.282
Multinomial	50	0.300	0.364	0.308
Multinomial	50	0.325	0.383	0.326
Multinomial	50	0.350	0.402	0.345
Multinomial	50	0.375	0.419	0.362
Multinomial	50	0.400	0.438	0.381
Multinomial	75	0.025	0.064	0.028
Multinomial	75	0.050	0.090	0.053
Multinomial	75	0.075	0.113	0.076
Multinomial	75	0.100	0.140	0.103
Multinomial	75	0.125	0.166	0.129

Table 1: Estimated R2 and adjusted R2 vs. population R2, by model and sample size. *(continued)*

Model	N	Population R2	R2	Adjusted R2
Multinomial	75	0.150	0.194	0.157
Multinomial	75	0.175	0.216	0.179
Multinomial	75	0.200	0.242	0.205
Multinomial	75	0.225	0.269	0.232
Multinomial	75	0.250	0.294	0.256
Multinomial	75	0.275	0.316	0.279
Multinomial	75	0.300	0.341	0.303
Multinomial	75	0.325	0.364	0.326
Multinomial	75	0.350	0.386	0.348
Multinomial	75	0.375	0.407	0.370
Multinomial	75	0.400	0.423	0.385
Multinomial	100	0.025	0.054	0.027
Multinomial	100	0.050	0.079	0.052
Multinomial	100	0.075	0.105	0.078
Multinomial	100	0.100	0.129	0.102
Multinomial	100	0.125	0.155	0.127
Multinomial	100	0.150	0.179	0.152
Multinomial	100	0.175	0.205	0.178
Multinomial	100	0.200	0.230	0.202
Multinomial	100	0.225	0.257	0.229
Multinomial	100	0.250	0.280	0.252
Multinomial	100	0.275	0.306	0.278
Multinomial	100	0.300	0.329	0.301
Multinomial	100	0.325	0.353	0.325
Multinomial	100	0.350	0.374	0.346
Multinomial	100	0.375	0.394	0.366
Multinomial	100	0.400	0.413	0.385
Ordinal	25	0.025	0.092	0.035
Ordinal	25	0.050	0.120	0.063
Ordinal	25	0.075	0.143	0.086
Ordinal	25	0.100	0.173	0.116
Ordinal	25	0.125	0.194	0.137
Ordinal	25	0.150	0.222	0.164
Ordinal	25	0.175	0.251	0.193
Ordinal	25	0.200	0.280	0.222
Ordinal	25	0.225	0.303	0.244
Ordinal	25	0.250	0.328	0.269
Ordinal	25	0.275	0.359	0.299
Ordinal	25	0.300	0.382	0.323
Ordinal	25	0.325	0.402	0.342
Ordinal	25	0.350	0.427	0.367
Ordinal	25	0.375	0.450	0.389
Ordinal	25	0.400	0.472	0.410
Ordinal	50	0.025	0.056	0.029
Ordinal	50	0.050	0.081	0.053
Ordinal	50	0.075	0.107	0.079
Ordinal	50	0.100	0.132	0.104
Ordinal	50	0.125	0.158	0.130
Ordinal	50	0.150	0.183	0.155
Ordinal	50	0.175	0.208	0.180
Ordinal	50	0.200	0.236	0.207
Ordinal	50	0.225	0.262	0.234
Ordinal	50	0.250	0.287	0.258
Ordinal	50	0.275	0.314	0.285
Ordinal	50	0.300	0.341	0.312

Table 1: Estimated R2 and adjusted R2 vs. population R2, by model and sample size. *(continued)*

Model	N	Population R2	R2	Adjusted R2
Ordinal	50	0.325	0.364	0.334
Ordinal	50	0.350	0.393	0.364
Ordinal	50	0.375	0.419	0.389
Ordinal	50	0.400	0.441	0.411
Ordinal	75	0.025	0.045	0.026
Ordinal	75	0.050	0.071	0.053
Ordinal	75	0.075	0.095	0.077
Ordinal	75	0.100	0.120	0.101
Ordinal	75	0.125	0.146	0.127
Ordinal	75	0.150	0.172	0.153
Ordinal	75	0.175	0.197	0.179
Ordinal	75	0.200	0.223	0.204
Ordinal	75	0.225	0.248	0.229
Ordinal	75	0.250	0.275	0.256
Ordinal	75	0.275	0.300	0.281
Ordinal	75	0.300	0.325	0.306
Ordinal	75	0.325	0.351	0.331
Ordinal	75	0.350	0.378	0.358
Ordinal	75	0.375	0.402	0.382
Ordinal	75	0.400	0.429	0.410
Ordinal	100	0.025	0.039	0.026
Ordinal	100	0.050	0.065	0.051
Ordinal	100	0.075	0.089	0.075
Ordinal	100	0.100	0.114	0.100
Ordinal	100	0.125	0.140	0.126
Ordinal	100	0.150	0.165	0.151
Ordinal	100	0.175	0.191	0.177
Ordinal	100	0.200	0.218	0.204
Ordinal	100	0.225	0.243	0.228
Ordinal	100	0.250	0.269	0.254
Ordinal	100	0.275	0.293	0.279
Ordinal	100	0.300	0.321	0.306
Ordinal	100	0.325	0.346	0.331
Ordinal	100	0.350	0.370	0.356
Ordinal	100	0.375	0.395	0.380
Ordinal	100	0.400	0.420	0.405

Study 1a: power of the omnibus test (data underlying Figure 3)

Table 2: Simulated power vs. closed-form power (from mean R2/adjusted R2), by model and sample size.

Model	N	Population R2	Power (Actual)	Power (R2)	Power (Adjusted R2)
Gaussian	25	0.025	0.088	0.324	0.083
Gaussian	25	0.050	0.120	0.370	0.117
Gaussian	25	0.075	0.167	0.423	0.160
Gaussian	25	0.100	0.219	0.479	0.209
Gaussian	25	0.125	0.265	0.532	0.260
Gaussian	25	0.150	0.321	0.581	0.310
Gaussian	25	0.175	0.380	0.635	0.371
Gaussian	25	0.200	0.448	0.688	0.435
Gaussian	25	0.225	0.500	0.728	0.486
Gaussian	25	0.250	0.551	0.771	0.544

Table 2: Simulated power vs. closed-form power (from mean R2/adjusted R2), by model and sample size. *(continued)*

Model	N	Population R2	Power (Actual)	Power (R2)	Power (Adjusted R2)
Gaussian	25	0.275	0.600	0.812	0.606
Gaussian	25	0.300	0.667	0.851	0.667
Gaussian	25	0.325	0.713	0.883	0.722
Gaussian	25	0.350	0.767	0.908	0.769
Gaussian	25	0.375	0.805	0.932	0.817
Gaussian	25	0.400	0.836	0.952	0.860
Gaussian	50	0.025	0.123	0.371	0.121
Gaussian	50	0.050	0.223	0.482	0.223
Gaussian	50	0.075	0.338	0.575	0.322
Gaussian	50	0.100	0.438	0.667	0.432
Gaussian	50	0.125	0.545	0.751	0.545
Gaussian	50	0.150	0.642	0.818	0.645
Gaussian	50	0.175	0.728	0.869	0.728
Gaussian	50	0.200	0.796	0.912	0.804
Gaussian	50	0.225	0.848	0.939	0.858
Gaussian	50	0.250	0.896	0.965	0.910
Gaussian	50	0.275	0.935	0.980	0.946
Gaussian	50	0.300	0.955	0.989	0.967
Gaussian	50	0.325	0.969	0.994	0.981
Gaussian	50	0.350	0.985	0.997	0.990
Gaussian	50	0.375	0.992	0.999	0.996
Gaussian	50	0.400	0.997	1.000	0.998
Gaussian	75	0.025	0.169	0.423	0.170
Gaussian	75	0.050	0.316	0.570	0.322
Gaussian	75	0.075	0.501	0.705	0.491
Gaussian	75	0.100	0.636	0.806	0.638
Gaussian	75	0.125	0.752	0.880	0.757
Gaussian	75	0.150	0.832	0.925	0.839
Gaussian	75	0.175	0.910	0.962	0.910
Gaussian	75	0.200	0.940	0.981	0.951
Gaussian	75	0.225	0.968	0.991	0.974
Gaussian	75	0.250	0.984	0.996	0.989
Gaussian	75	0.275	0.994	0.999	0.995
Gaussian	75	0.300	0.998	1.000	0.998
Gaussian	75	0.325	0.997	1.000	0.999
Gaussian	75	0.350	1.000	1.000	1.000
Gaussian	75	0.375	1.000	1.000	1.000
Gaussian	75	0.400	1.000	1.000	1.000
Gaussian	100	0.025	0.230	0.483	0.230
Gaussian	100	0.050	0.447	0.668	0.446
Gaussian	100	0.075	0.636	0.798	0.630
Gaussian	100	0.100	0.791	0.897	0.791
Gaussian	100	0.125	0.883	0.946	0.882
Gaussian	100	0.150	0.944	0.977	0.946
Gaussian	100	0.175	0.968	0.990	0.976
Gaussian	100	0.200	0.987	0.996	0.990
Gaussian	100	0.225	0.993	0.999	0.996
Gaussian	100	0.250	0.997	1.000	0.999
Gaussian	100	0.275	0.999	1.000	1.000
Gaussian	100	0.300	0.999	1.000	1.000
Gaussian	100	0.325	1.000	1.000	1.000
Gaussian	100	0.350	1.000	1.000	1.000
Gaussian	100	0.375	1.000	1.000	1.000
Gaussian	100	0.400	1.000	1.000	1.000
Logistic	25	0.025	0.136	0.393	0.139

Table 2: Simulated power vs. closed-form power (from mean R2/adjusted R2), by model and sample size. *(continued)*

Model	N	Population R2	Power (Actual)	Power (R2)	Power (Adjusted R2)
Logistic	25	0.050	0.215	0.472	0.217
Logistic	25	0.075	0.281	0.536	0.286
Logistic	25	0.100	0.348	0.598	0.361
Logistic	25	0.125	0.430	0.659	0.439
Logistic	25	0.150	0.521	0.711	0.511
Logistic	25	0.175	0.575	0.749	0.565
Logistic	25	0.200	0.634	0.786	0.620
Logistic	25	0.225	0.690	0.815	0.666
Logistic	25	0.250	0.746	0.842	0.709
Logistic	25	0.275	0.795	0.871	0.757
Logistic	25	0.300	0.830	0.889	0.788
Logistic	25	0.325	0.870	0.906	0.818
Logistic	25	0.350	0.902	0.921	0.845
Logistic	25	0.375	0.923	0.932	0.863
Logistic	25	0.400	0.938	0.941	0.881
Logistic	50	0.025	0.184	0.441	0.189
Logistic	50	0.050	0.324	0.567	0.327
Logistic	50	0.075	0.485	0.686	0.479
Logistic	50	0.100	0.610	0.771	0.602
Logistic	50	0.125	0.722	0.843	0.713
Logistic	50	0.150	0.810	0.894	0.798
Logistic	50	0.175	0.878	0.928	0.858
Logistic	50	0.200	0.924	0.953	0.904
Logistic	50	0.225	0.953	0.968	0.934
Logistic	50	0.250	0.974	0.979	0.956
Logistic	50	0.275	0.980	0.987	0.972
Logistic	50	0.300	0.994	0.993	0.983
Logistic	50	0.325	0.997	0.995	0.989
Logistic	50	0.350	0.998	0.997	0.993
Logistic	50	0.375	0.999	0.998	0.996
Logistic	50	0.400	1.000	0.999	0.997
Logistic	75	0.025	0.246	0.501	0.254
Logistic	75	0.050	0.474	0.680	0.473
Logistic	75	0.075	0.669	0.806	0.656
Logistic	75	0.100	0.806	0.890	0.792
Logistic	75	0.125	0.890	0.938	0.877
Logistic	75	0.150	0.942	0.967	0.931
Logistic	75	0.175	0.974	0.983	0.963
Logistic	75	0.200	0.990	0.992	0.982
Logistic	75	0.225	0.996	0.996	0.991
Logistic	75	0.250	0.999	0.998	0.995
Logistic	75	0.275	1.000	0.999	0.998
Logistic	75	0.300	1.000	1.000	0.999
Logistic	75	0.325	1.000	1.000	1.000
Logistic	75	0.350	1.000	1.000	1.000
Logistic	75	0.375	1.000	1.000	1.000
Logistic	75	0.400	1.000	1.000	1.000
Logistic	100	0.025	0.317	0.560	0.321
Logistic	100	0.050	0.600	0.768	0.598
Logistic	100	0.075	0.802	0.889	0.791
Logistic	100	0.100	0.912	0.949	0.898
Logistic	100	0.125	0.964	0.978	0.953
Logistic	100	0.150	0.986	0.991	0.980
Logistic	100	0.175	0.996	0.997	0.992
Logistic	100	0.200	0.998	0.999	0.997

Table 2: Simulated power vs. closed-form power (from mean R2/adjusted R2), by model and sample size. *(continued)*

Model	N	Population R2	Power (Actual)	Power (R2)	Power (Adjusted R2)
Logistic	100	0.225	0.999	1.000	0.999
Logistic	100	0.250	1.000	1.000	1.000
Logistic	100	0.275	1.000	1.000	1.000
Logistic	100	0.300	1.000	1.000	1.000
Logistic	100	0.325	1.000	1.000	1.000
Logistic	100	0.350	1.000	1.000	1.000
Logistic	100	0.375	1.000	1.000	1.000
Logistic	100	0.400	1.000	1.000	1.000
Multinomial	25	0.025	0.177	0.701	0.240
Multinomial	25	0.050	0.268	0.774	0.362
Multinomial	25	0.075	0.355	0.828	0.473
Multinomial	25	0.100	0.428	0.866	0.561
Multinomial	25	0.125	0.542	0.904	0.661
Multinomial	25	0.150	0.610	0.925	0.720
Multinomial	25	0.175	0.699	0.945	0.783
Multinomial	25	0.200	0.754	0.958	0.828
Multinomial	25	0.225	0.814	0.968	0.861
Multinomial	25	0.250	0.862	0.976	0.891
Multinomial	25	0.275	0.907	0.981	0.913
Multinomial	25	0.300	0.929	0.985	0.930
Multinomial	25	0.325	0.952	0.988	0.940
Multinomial	25	0.350	0.969	0.990	0.951
Multinomial	25	0.375	0.985	0.992	0.958
Multinomial	25	0.400	0.984	0.993	0.964
Multinomial	50	0.025	0.220	0.737	0.306
Multinomial	50	0.050	0.409	0.851	0.534
Multinomial	50	0.075	0.597	0.920	0.715
Multinomial	50	0.100	0.735	0.958	0.831
Multinomial	50	0.125	0.846	0.980	0.912
Multinomial	50	0.150	0.922	0.990	0.952
Multinomial	50	0.175	0.959	0.996	0.977
Multinomial	50	0.200	0.983	0.998	0.988
Multinomial	50	0.225	0.992	0.999	0.995
Multinomial	50	0.250	0.998	1.000	0.997
Multinomial	50	0.275	0.999	1.000	0.999
Multinomial	50	0.300	1.000	1.000	0.999
Multinomial	50	0.325	1.000	1.000	1.000
Multinomial	50	0.350	1.000	1.000	1.000
Multinomial	50	0.375	1.000	1.000	1.000
Multinomial	50	0.400	1.000	1.000	1.000
Multinomial	75	0.025	0.296	0.788	0.403
Multinomial	75	0.050	0.567	0.913	0.694
Multinomial	75	0.075	0.773	0.965	0.859
Multinomial	75	0.100	0.912	0.989	0.948
Multinomial	75	0.125	0.970	0.997	0.982
Multinomial	75	0.150	0.993	0.999	0.995
Multinomial	75	0.175	0.996	1.000	0.998
Multinomial	75	0.200	1.000	1.000	0.999
Multinomial	75	0.225	1.000	1.000	1.000
Multinomial	75	0.250	1.000	1.000	1.000
Multinomial	75	0.275	1.000	1.000	1.000
Multinomial	75	0.300	1.000	1.000	1.000
Multinomial	75	0.325	1.000	1.000	1.000
Multinomial	75	0.350	1.000	1.000	1.000
Multinomial	75	0.375	1.000	1.000	1.000

Table 2: Simulated power vs. closed-form power (from mean R2/adjusted R2), by model and sample size. *(continued)*

Model	N	Population R2	Power (Actual)	Power (R2)	Power (Adjusted R2)
Multinomial	75	0.400	1.000	1.000	1.000
Multinomial	100	0.025	0.381	0.837	0.506
Multinomial	100	0.050	0.715	0.953	0.818
Multinomial	100	0.075	0.899	0.989	0.949
Multinomial	100	0.100	0.972	0.997	0.986
Multinomial	100	0.125	0.993	0.999	0.997
Multinomial	100	0.150	0.998	1.000	0.999
Multinomial	100	0.175	1.000	1.000	1.000
Multinomial	100	0.200	1.000	1.000	1.000
Multinomial	100	0.225	1.000	1.000	1.000
Multinomial	100	0.250	1.000	1.000	1.000
Multinomial	100	0.275	1.000	1.000	1.000
Multinomial	100	0.300	1.000	1.000	1.000
Multinomial	100	0.325	1.000	1.000	1.000
Multinomial	100	0.350	1.000	1.000	1.000
Multinomial	100	0.375	1.000	1.000	1.000
Multinomial	100	0.400	1.000	1.000	1.000
Ordinal	25	0.025	0.180	0.443	0.185
Ordinal	25	0.050	0.305	0.560	0.313
Ordinal	25	0.075	0.409	0.646	0.419
Ordinal	25	0.100	0.528	0.737	0.545
Ordinal	25	0.125	0.612	0.791	0.625
Ordinal	25	0.150	0.713	0.846	0.713
Ordinal	25	0.175	0.786	0.891	0.788
Ordinal	25	0.200	0.850	0.924	0.846
Ordinal	25	0.225	0.888	0.943	0.882
Ordinal	25	0.250	0.922	0.959	0.913
Ordinal	25	0.275	0.957	0.973	0.941
Ordinal	25	0.300	0.969	0.981	0.956
Ordinal	25	0.325	0.980	0.985	0.966
Ordinal	25	0.350	0.991	0.990	0.976
Ordinal	25	0.375	0.989	0.993	0.982
Ordinal	25	0.400	0.996	0.995	0.987
Ordinal	50	0.025	0.281	0.532	0.286
Ordinal	50	0.050	0.513	0.707	0.507
Ordinal	50	0.075	0.699	0.833	0.696
Ordinal	50	0.100	0.823	0.906	0.818
Ordinal	50	0.125	0.911	0.952	0.902
Ordinal	50	0.150	0.957	0.976	0.948
Ordinal	50	0.175	0.979	0.988	0.974
Ordinal	50	0.200	0.991	0.995	0.988
Ordinal	50	0.225	0.997	0.998	0.994
Ordinal	50	0.250	0.999	0.999	0.997
Ordinal	50	0.275	0.999	1.000	0.999
Ordinal	50	0.300	1.000	1.000	1.000
Ordinal	50	0.325	1.000	1.000	1.000
Ordinal	50	0.350	1.000	1.000	1.000
Ordinal	50	0.375	1.000	1.000	1.000
Ordinal	50	0.400	1.000	1.000	1.000
Ordinal	75	0.025	0.379	0.613	0.384
Ordinal	75	0.050	0.691	0.831	0.694
Ordinal	75	0.075	0.862	0.930	0.862
Ordinal	75	0.100	0.952	0.974	0.944
Ordinal	75	0.125	0.981	0.991	0.980
Ordinal	75	0.150	0.995	0.997	0.994

Table 2: Simulated power vs. closed-form power (from mean R2/adjusted R2), by model and sample size. *(continued)*

Model	N	Population R2	Power (Actual)	Power (R2)	Power (Adjusted R2)
Ordinal	75	0.175	0.999	0.999	0.998
Ordinal	75	0.200	1.000	1.000	0.999
Ordinal	75	0.225	1.000	1.000	1.000
Ordinal	75	0.250	1.000	1.000	1.000
Ordinal	75	0.275	1.000	1.000	1.000
Ordinal	75	0.300	1.000	1.000	1.000
Ordinal	75	0.325	1.000	1.000	1.000
Ordinal	75	0.350	1.000	1.000	1.000
Ordinal	75	0.375	1.000	1.000	1.000
Ordinal	75	0.400	1.000	1.000	1.000
Ordinal	100	0.025	0.486	0.691	0.488
Ordinal	100	0.050	0.813	0.901	0.811
Ordinal	100	0.075	0.947	0.972	0.942
Ordinal	100	0.100	0.990	0.993	0.985
Ordinal	100	0.125	0.998	0.999	0.997
Ordinal	100	0.150	0.999	1.000	0.999
Ordinal	100	0.175	1.000	1.000	1.000
Ordinal	100	0.200	1.000	1.000	1.000
Ordinal	100	0.225	1.000	1.000	1.000
Ordinal	100	0.250	1.000	1.000	1.000
Ordinal	100	0.275	1.000	1.000	1.000
Ordinal	100	0.300	1.000	1.000	1.000
Ordinal	100	0.325	1.000	1.000	1.000
Ordinal	100	0.350	1.000	1.000	1.000
Ordinal	100	0.375	1.000	1.000	1.000
Ordinal	100	0.400	1.000	1.000	1.000

Study 1b: accuracy of eta-squared (data underlying Figure 2)

Table 3: Estimated eta-squared and adjusted epsilon-squared vs. population eta-squared, by model and sample size.

Model	N	Population Eta2	Eta2	Epsilon2
Gaussian	25	0.025	0.061	0.036
Gaussian	25	0.050	0.079	0.052
Gaussian	25	0.075	0.101	0.072
Gaussian	25	0.100	0.120	0.091
Gaussian	25	0.125	0.143	0.113
Gaussian	25	0.150	0.162	0.131
Gaussian	25	0.175	0.183	0.153
Gaussian	25	0.200	0.207	0.177
Gaussian	25	0.225	0.225	0.195
Gaussian	25	0.250	0.247	0.218
Gaussian	25	0.275	0.273	0.245
Gaussian	25	0.300	0.289	0.262
Gaussian	25	0.325	0.310	0.284
Gaussian	25	0.350	0.344	0.319
Gaussian	25	0.375	0.355	0.331
Gaussian	25	0.400	0.374	0.351
Gaussian	50	0.025	0.042	0.029
Gaussian	50	0.050	0.065	0.049
Gaussian	50	0.075	0.088	0.071
Gaussian	50	0.100	0.111	0.094

Table 3: Estimated eta-squared and adjusted epsilon-squared vs. population eta-squared, by model and sample size. *(continued)*

Model	N	Population Eta2	Eta2	Epsilon2
Gaussian	50	0.125	0.128	0.111
Gaussian	50	0.150	0.154	0.138
Gaussian	50	0.175	0.178	0.162
Gaussian	50	0.200	0.199	0.183
Gaussian	50	0.225	0.230	0.215
Gaussian	50	0.250	0.252	0.237
Gaussian	50	0.275	0.267	0.253
Gaussian	50	0.300	0.291	0.277
Gaussian	50	0.325	0.320	0.307
Gaussian	50	0.350	0.342	0.329
Gaussian	50	0.375	0.365	0.354
Gaussian	50	0.400	0.389	0.378
Gaussian	75	0.025	0.038	0.028
Gaussian	75	0.050	0.058	0.047
Gaussian	75	0.075	0.087	0.075
Gaussian	75	0.100	0.105	0.093
Gaussian	75	0.125	0.130	0.118
Gaussian	75	0.150	0.156	0.144
Gaussian	75	0.175	0.181	0.170
Gaussian	75	0.200	0.202	0.191
Gaussian	75	0.225	0.230	0.220
Gaussian	75	0.250	0.249	0.239
Gaussian	75	0.275	0.273	0.264
Gaussian	75	0.300	0.295	0.286
Gaussian	75	0.325	0.314	0.306
Gaussian	75	0.350	0.349	0.341
Gaussian	75	0.375	0.366	0.358
Gaussian	75	0.400	0.390	0.383
Gaussian	100	0.025	0.034	0.026
Gaussian	100	0.050	0.057	0.048
Gaussian	100	0.075	0.084	0.075
Gaussian	100	0.100	0.105	0.096
Gaussian	100	0.125	0.128	0.120
Gaussian	100	0.150	0.151	0.143
Gaussian	100	0.175	0.176	0.168
Gaussian	100	0.200	0.201	0.193
Gaussian	100	0.225	0.226	0.219
Gaussian	100	0.250	0.252	0.245
Gaussian	100	0.275	0.274	0.267
Gaussian	100	0.300	0.298	0.292
Gaussian	100	0.325	0.323	0.316
Gaussian	100	0.350	0.343	0.337
Gaussian	100	0.375	0.370	0.365
Gaussian	100	0.400	0.391	0.386
Logistic	25	0.025	0.062	0.042
Logistic	25	0.050	0.089	0.067
Logistic	25	0.075	0.109	0.084
Logistic	25	0.100	0.136	0.110
Logistic	25	0.125	0.161	0.134
Logistic	25	0.150	0.184	0.156
Logistic	25	0.175	0.212	0.183
Logistic	25	0.200	0.235	0.206
Logistic	25	0.225	0.260	0.230
Logistic	25	0.250	0.284	0.254
Logistic	25	0.275	0.304	0.274

Table 3: Estimated eta-squared and adjusted epsilon-squared vs. population eta-squared, by model and sample size. *(continued)*

Model	N	Population Eta2	Eta2	Epsilon2
Logistic	25	0.300	0.326	0.296
Logistic	25	0.325	0.338	0.308
Logistic	25	0.350	0.360	0.330
Logistic	25	0.375	0.381	0.351
Logistic	25	0.400	0.393	0.364
Logistic	50	0.025	0.044	0.033
Logistic	50	0.050	0.069	0.056
Logistic	50	0.075	0.092	0.078
Logistic	50	0.100	0.119	0.105
Logistic	50	0.125	0.143	0.129
Logistic	50	0.150	0.168	0.153
Logistic	50	0.175	0.194	0.179
Logistic	50	0.200	0.220	0.205
Logistic	50	0.225	0.240	0.225
Logistic	50	0.250	0.268	0.254
Logistic	50	0.275	0.297	0.282
Logistic	50	0.300	0.318	0.304
Logistic	50	0.325	0.347	0.332
Logistic	50	0.350	0.366	0.351
Logistic	50	0.375	0.396	0.381
Logistic	50	0.400	0.415	0.400
Logistic	75	0.025	0.035	0.027
Logistic	75	0.050	0.062	0.053
Logistic	75	0.075	0.087	0.078
Logistic	75	0.100	0.111	0.101
Logistic	75	0.125	0.139	0.129
Logistic	75	0.150	0.162	0.153
Logistic	75	0.175	0.192	0.183
Logistic	75	0.200	0.218	0.208
Logistic	75	0.225	0.238	0.228
Logistic	75	0.250	0.261	0.251
Logistic	75	0.275	0.289	0.279
Logistic	75	0.300	0.310	0.300
Logistic	75	0.325	0.347	0.337
Logistic	75	0.350	0.362	0.352
Logistic	75	0.375	0.385	0.376
Logistic	75	0.400	0.411	0.401
Logistic	100	0.025	0.033	0.027
Logistic	100	0.050	0.059	0.052
Logistic	100	0.075	0.085	0.078
Logistic	100	0.100	0.109	0.101
Logistic	100	0.125	0.131	0.124
Logistic	100	0.150	0.160	0.153
Logistic	100	0.175	0.180	0.173
Logistic	100	0.200	0.206	0.198
Logistic	100	0.225	0.232	0.225
Logistic	100	0.250	0.260	0.253
Logistic	100	0.275	0.282	0.275
Logistic	100	0.300	0.306	0.299
Logistic	100	0.325	0.331	0.324
Logistic	100	0.350	0.357	0.350
Logistic	100	0.375	0.381	0.374
Logistic	100	0.400	0.404	0.397
Multinomial	25	0.025	0.073	0.043
Multinomial	25	0.050	0.101	0.068

Table 3: Estimated eta-squared and adjusted epsilon-squared vs. population eta-squared, by model and sample size. *(continued)*

Model	N	Population Eta2	Eta2	Epsilon2
Multinomial	25	0.075	0.122	0.087
Multinomial	25	0.100	0.151	0.115
Multinomial	25	0.125	0.176	0.139
Multinomial	25	0.150	0.190	0.153
Multinomial	25	0.175	0.213	0.176
Multinomial	25	0.200	0.232	0.194
Multinomial	25	0.225	0.257	0.219
Multinomial	25	0.250	0.275	0.237
Multinomial	25	0.275	0.286	0.248
Multinomial	25	0.300	0.306	0.268
Multinomial	25	0.325	0.313	0.274
Multinomial	25	0.350	0.323	0.284
Multinomial	25	0.375	0.338	0.300
Multinomial	25	0.400	0.354	0.315
Multinomial	50	0.025	0.046	0.030
Multinomial	50	0.050	0.073	0.055
Multinomial	50	0.075	0.097	0.079
Multinomial	50	0.100	0.122	0.103
Multinomial	50	0.125	0.147	0.128
Multinomial	50	0.150	0.179	0.160
Multinomial	50	0.175	0.199	0.180
Multinomial	50	0.200	0.223	0.204
Multinomial	50	0.225	0.248	0.229
Multinomial	50	0.250	0.276	0.257
Multinomial	50	0.275	0.296	0.277
Multinomial	50	0.300	0.317	0.298
Multinomial	50	0.325	0.339	0.320
Multinomial	50	0.350	0.350	0.331
Multinomial	50	0.375	0.369	0.350
Multinomial	50	0.400	0.382	0.363
Multinomial	75	0.025	0.040	0.028
Multinomial	75	0.050	0.065	0.053
Multinomial	75	0.075	0.089	0.077
Multinomial	75	0.100	0.114	0.101
Multinomial	75	0.125	0.141	0.129
Multinomial	75	0.150	0.163	0.151
Multinomial	75	0.175	0.191	0.178
Multinomial	75	0.200	0.217	0.204
Multinomial	75	0.225	0.239	0.227
Multinomial	75	0.250	0.264	0.251
Multinomial	75	0.275	0.290	0.277
Multinomial	75	0.300	0.311	0.298
Multinomial	75	0.325	0.332	0.320
Multinomial	75	0.350	0.353	0.341
Multinomial	75	0.375	0.366	0.354
Multinomial	75	0.400	0.383	0.371
Multinomial	100	0.025	0.036	0.027
Multinomial	100	0.050	0.059	0.050
Multinomial	100	0.075	0.085	0.076
Multinomial	100	0.100	0.109	0.100
Multinomial	100	0.125	0.133	0.124
Multinomial	100	0.150	0.164	0.155
Multinomial	100	0.175	0.183	0.174
Multinomial	100	0.200	0.213	0.204
Multinomial	100	0.225	0.237	0.227

Table 3: Estimated eta-squared and adjusted epsilon-squared vs. population eta-squared, by model and sample size. *(continued)*

Model	N	Population Eta2	Eta2	Epsilon2
Multinomial	100	0.250	0.260	0.251
Multinomial	100	0.275	0.282	0.273
Multinomial	100	0.300	0.313	0.304
Multinomial	100	0.325	0.327	0.317
Multinomial	100	0.350	0.350	0.340
Multinomial	100	0.375	0.367	0.358
Multinomial	100	0.400	0.382	0.373
Ordinal	25	0.025	0.051	0.037
Ordinal	25	0.050	0.079	0.062
Ordinal	25	0.075	0.100	0.082
Ordinal	25	0.100	0.129	0.111
Ordinal	25	0.125	0.158	0.140
Ordinal	25	0.150	0.180	0.161
Ordinal	25	0.175	0.202	0.183
Ordinal	25	0.200	0.237	0.217
Ordinal	25	0.225	0.262	0.243
Ordinal	25	0.250	0.289	0.270
Ordinal	25	0.275	0.309	0.289
Ordinal	25	0.300	0.338	0.318
Ordinal	25	0.325	0.360	0.340
Ordinal	25	0.350	0.386	0.366
Ordinal	25	0.375	0.402	0.382
Ordinal	25	0.400	0.430	0.410
Ordinal	50	0.025	0.034	0.026
Ordinal	50	0.050	0.061	0.052
Ordinal	50	0.075	0.088	0.078
Ordinal	50	0.100	0.110	0.101
Ordinal	50	0.125	0.138	0.129
Ordinal	50	0.150	0.160	0.151
Ordinal	50	0.175	0.185	0.176
Ordinal	50	0.200	0.219	0.209
Ordinal	50	0.225	0.234	0.224
Ordinal	50	0.250	0.265	0.255
Ordinal	50	0.275	0.289	0.279
Ordinal	50	0.300	0.320	0.310
Ordinal	50	0.325	0.348	0.339
Ordinal	50	0.350	0.370	0.360
Ordinal	50	0.375	0.396	0.386
Ordinal	50	0.400	0.422	0.412
Ordinal	75	0.025	0.033	0.028
Ordinal	75	0.050	0.056	0.050
Ordinal	75	0.075	0.083	0.077
Ordinal	75	0.100	0.109	0.103
Ordinal	75	0.125	0.133	0.127
Ordinal	75	0.150	0.160	0.154
Ordinal	75	0.175	0.188	0.182
Ordinal	75	0.200	0.210	0.204
Ordinal	75	0.225	0.233	0.227
Ordinal	75	0.250	0.261	0.254
Ordinal	75	0.275	0.288	0.282
Ordinal	75	0.300	0.315	0.308
Ordinal	75	0.325	0.341	0.334
Ordinal	75	0.350	0.365	0.359
Ordinal	75	0.375	0.383	0.376
Ordinal	75	0.400	0.413	0.406

Table 3: Estimated eta-squared and adjusted epsilon-squared vs. population eta-squared, by model and sample size. *(continued)*

Model	N	Population Eta2	Eta2	Epsilon2
Ordinal	100	0.025	0.029	0.025
Ordinal	100	0.050	0.057	0.053
Ordinal	100	0.075	0.080	0.076
Ordinal	100	0.100	0.105	0.100
Ordinal	100	0.125	0.132	0.127
Ordinal	100	0.150	0.156	0.151
Ordinal	100	0.175	0.180	0.176
Ordinal	100	0.200	0.209	0.205
Ordinal	100	0.225	0.231	0.226
Ordinal	100	0.250	0.257	0.252
Ordinal	100	0.275	0.278	0.273
Ordinal	100	0.300	0.310	0.305
Ordinal	100	0.325	0.335	0.330
Ordinal	100	0.350	0.360	0.355
Ordinal	100	0.375	0.383	0.378
Ordinal	100	0.400	0.409	0.404

Study 1b: power of the single-predictor test (data underlying Figure 4)

Table 4: Simulated power vs. closed-form power (from mean eta-squared/epsilon-squared), by model and sample size.

Model	N	Population Eta2	Power (Actual)	Power (Eta2)	Power (Epsilon2)
Gaussian	25	0.025	0.102	0.215	0.145
Gaussian	25	0.050	0.170	0.268	0.188
Gaussian	25	0.075	0.243	0.336	0.249
Gaussian	25	0.100	0.316	0.395	0.305
Gaussian	25	0.125	0.408	0.465	0.373
Gaussian	25	0.150	0.464	0.520	0.428
Gaussian	25	0.175	0.527	0.581	0.493
Gaussian	25	0.200	0.639	0.648	0.565
Gaussian	25	0.225	0.666	0.692	0.614
Gaussian	25	0.250	0.722	0.745	0.674
Gaussian	25	0.275	0.794	0.800	0.740
Gaussian	25	0.300	0.821	0.830	0.778
Gaussian	25	0.325	0.873	0.866	0.821
Gaussian	25	0.350	0.904	0.912	0.879
Gaussian	25	0.375	0.937	0.924	0.895
Gaussian	25	0.400	0.950	0.942	0.920
Gaussian	50	0.025	0.177	0.297	0.215
Gaussian	50	0.050	0.327	0.430	0.337
Gaussian	50	0.075	0.504	0.557	0.467
Gaussian	50	0.100	0.605	0.668	0.588
Gaussian	50	0.125	0.686	0.740	0.671
Gaussian	50	0.150	0.782	0.826	0.774
Gaussian	50	0.175	0.870	0.885	0.847
Gaussian	50	0.200	0.910	0.922	0.894
Gaussian	50	0.225	0.953	0.960	0.944
Gaussian	50	0.250	0.967	0.976	0.966
Gaussian	50	0.275	0.981	0.984	0.977
Gaussian	50	0.300	0.988	0.991	0.988
Gaussian	50	0.325	0.996	0.996	0.995
Gaussian	50	0.350	0.998	0.998	0.997

Table 4: Simulated power vs. closed-form power (from mean eta-squared/epsilon-squared), by model and sample size. *(continued)*

Model	N	Population Eta2	Power (Actual)	Power (Eta2)	Power (Epsilon2)
Gaussian	50	0.375	1.000	0.999	0.999
Gaussian	50	0.400	1.000	1.000	1.000
Gaussian	75	0.025	0.274	0.384	0.294
Gaussian	75	0.050	0.481	0.552	0.463
Gaussian	75	0.075	0.682	0.738	0.669
Gaussian	75	0.100	0.782	0.821	0.769
Gaussian	75	0.125	0.876	0.902	0.869
Gaussian	75	0.150	0.932	0.951	0.934
Gaussian	75	0.175	0.966	0.977	0.968
Gaussian	75	0.200	0.984	0.989	0.984
Gaussian	75	0.225	0.993	0.996	0.994
Gaussian	75	0.250	0.997	0.998	0.997
Gaussian	75	0.275	0.998	0.999	0.999
Gaussian	75	0.300	1.000	1.000	1.000
Gaussian	75	0.325	0.999	1.000	1.000
Gaussian	75	0.350	1.000	1.000	1.000
Gaussian	75	0.375	1.000	1.000	1.000
Gaussian	75	0.400	1.000	1.000	1.000
Gaussian	100	0.025	0.349	0.453	0.361
Gaussian	100	0.050	0.608	0.670	0.591
Gaussian	100	0.075	0.806	0.841	0.794
Gaussian	100	0.100	0.906	0.918	0.891
Gaussian	100	0.125	0.964	0.964	0.951
Gaussian	100	0.150	0.988	0.985	0.979
Gaussian	100	0.175	0.991	0.995	0.993
Gaussian	100	0.200	0.996	0.998	0.998
Gaussian	100	0.225	0.999	1.000	0.999
Gaussian	100	0.250	1.000	1.000	1.000
Gaussian	100	0.275	1.000	1.000	1.000
Gaussian	100	0.300	1.000	1.000	1.000
Gaussian	100	0.325	1.000	1.000	1.000
Gaussian	100	0.350	1.000	1.000	1.000
Gaussian	100	0.375	1.000	1.000	1.000
Gaussian	100	0.400	1.000	1.000	1.000
Logistic	25	0.025	0.183	0.312	0.227
Logistic	25	0.050	0.301	0.421	0.330
Logistic	25	0.075	0.363	0.494	0.400
Logistic	25	0.100	0.480	0.584	0.496
Logistic	25	0.125	0.563	0.657	0.578
Logistic	25	0.150	0.619	0.714	0.642
Logistic	25	0.175	0.716	0.774	0.713
Logistic	25	0.200	0.777	0.814	0.762
Logistic	25	0.225	0.829	0.851	0.807
Logistic	25	0.250	0.853	0.880	0.843
Logistic	25	0.275	0.887	0.900	0.869
Logistic	25	0.300	0.915	0.919	0.894
Logistic	25	0.325	0.922	0.928	0.905
Logistic	25	0.350	0.962	0.942	0.923
Logistic	25	0.375	0.964	0.953	0.937
Logistic	25	0.400	0.975	0.958	0.944
Logistic	50	0.025	0.287	0.415	0.325
Logistic	50	0.050	0.487	0.587	0.501
Logistic	50	0.075	0.650	0.715	0.644
Logistic	50	0.100	0.769	0.820	0.769
Logistic	50	0.125	0.848	0.883	0.847

Table 4: Simulated power vs. closed-form power (from mean eta-squared/epsilon-squared), by model and sample size. *(continued)*

Model	N	Population Eta2	Power (Actual)	Power (Eta2)	Power (Epsilon2)
Logistic	50	0.150	0.911	0.927	0.903
Logistic	50	0.175	0.951	0.956	0.941
Logistic	50	0.200	0.977	0.974	0.965
Logistic	50	0.225	0.977	0.983	0.977
Logistic	50	0.250	0.995	0.991	0.987
Logistic	50	0.275	0.997	0.995	0.993
Logistic	50	0.300	0.998	0.997	0.996
Logistic	50	0.325	0.999	0.998	0.998
Logistic	50	0.350	1.000	0.999	0.999
Logistic	50	0.375	1.000	0.999	0.999
Logistic	50	0.400	1.000	1.000	1.000
Logistic	75	0.025	0.366	0.483	0.391
Logistic	75	0.050	0.648	0.719	0.650
Logistic	75	0.075	0.816	0.854	0.812
Logistic	75	0.100	0.916	0.924	0.900
Logistic	75	0.125	0.951	0.967	0.956
Logistic	75	0.150	0.982	0.984	0.979
Logistic	75	0.175	0.991	0.994	0.992
Logistic	75	0.200	0.993	0.997	0.996
Logistic	75	0.225	1.000	0.999	0.998
Logistic	75	0.250	0.999	0.999	0.999
Logistic	75	0.275	0.999	1.000	1.000
Logistic	75	0.300	1.000	1.000	1.000
Logistic	75	0.325	1.000	1.000	1.000
Logistic	75	0.350	1.000	1.000	1.000
Logistic	75	0.375	1.000	1.000	1.000
Logistic	75	0.400	1.000	1.000	1.000
Logistic	100	0.025	0.490	0.571	0.485
Logistic	100	0.050	0.759	0.815	0.764
Logistic	100	0.075	0.909	0.929	0.907
Logistic	100	0.100	0.970	0.973	0.963
Logistic	100	0.125	0.987	0.990	0.986
Logistic	100	0.150	0.999	0.997	0.996
Logistic	100	0.175	1.000	0.999	0.998
Logistic	100	0.200	1.000	1.000	0.999
Logistic	100	0.225	1.000	1.000	1.000
Logistic	100	0.250	1.000	1.000	1.000
Logistic	100	0.275	1.000	1.000	1.000
Logistic	100	0.300	1.000	1.000	1.000
Logistic	100	0.325	1.000	1.000	1.000
Logistic	100	0.350	1.000	1.000	1.000
Logistic	100	0.375	1.000	1.000	1.000
Logistic	100	0.400	1.000	1.000	1.000
Multinomial	25	0.025	0.216	0.416	0.259
Multinomial	25	0.050	0.373	0.550	0.389
Multinomial	25	0.075	0.473	0.635	0.482
Multinomial	25	0.100	0.604	0.734	0.606
Multinomial	25	0.125	0.708	0.801	0.695
Multinomial	25	0.150	0.744	0.833	0.739
Multinomial	25	0.175	0.821	0.875	0.801
Multinomial	25	0.200	0.870	0.902	0.840
Multinomial	25	0.225	0.909	0.930	0.884
Multinomial	25	0.250	0.948	0.946	0.908
Multinomial	25	0.275	0.953	0.953	0.921
Multinomial	25	0.300	0.969	0.965	0.940

Table 4: Simulated power vs. closed-form power (from mean eta-squared/epsilon-squared), by model and sample size. *(continued)*

Model	N	Population Eta2	Power (Actual)	Power (Eta2)	Power (Epsilon2)
Multinomial	25	0.325	0.981	0.968	0.945
Multinomial	25	0.350	0.972	0.973	0.952
Multinomial	25	0.375	0.993	0.978	0.962
Multinomial	25	0.400	0.995	0.983	0.969
Multinomial	50	0.025	0.313	0.506	0.347
Multinomial	50	0.050	0.556	0.717	0.588
Multinomial	50	0.075	0.759	0.842	0.755
Multinomial	50	0.100	0.872	0.916	0.864
Multinomial	50	0.125	0.923	0.958	0.930
Multinomial	50	0.150	0.980	0.984	0.971
Multinomial	50	0.175	0.987	0.991	0.984
Multinomial	50	0.200	0.998	0.996	0.993
Multinomial	50	0.225	0.999	0.998	0.997
Multinomial	50	0.250	1.000	0.999	0.999
Multinomial	50	0.275	1.000	1.000	0.999
Multinomial	50	0.300	1.000	1.000	1.000
Multinomial	50	0.325	1.000	1.000	1.000
Multinomial	50	0.350	1.000	1.000	1.000
Multinomial	50	0.375	1.000	1.000	1.000
Multinomial	50	0.400	1.000	1.000	1.000
Multinomial	75	0.025	0.451	0.624	0.476
Multinomial	75	0.050	0.754	0.845	0.759
Multinomial	75	0.075	0.915	0.940	0.902
Multinomial	75	0.100	0.971	0.979	0.964
Multinomial	75	0.125	0.994	0.994	0.990
Multinomial	75	0.150	0.996	0.998	0.996
Multinomial	75	0.175	1.000	0.999	0.999
Multinomial	75	0.200	1.000	1.000	1.000
Multinomial	75	0.225	1.000	1.000	1.000
Multinomial	75	0.250	1.000	1.000	1.000
Multinomial	75	0.275	1.000	1.000	1.000
Multinomial	75	0.300	1.000	1.000	1.000
Multinomial	75	0.325	1.000	1.000	1.000
Multinomial	75	0.350	1.000	1.000	1.000
Multinomial	75	0.375	1.000	1.000	1.000
Multinomial	75	0.400	1.000	1.000	1.000
Multinomial	100	0.025	0.551	0.709	0.579
Multinomial	100	0.050	0.864	0.909	0.854
Multinomial	100	0.075	0.967	0.978	0.963
Multinomial	100	0.100	0.994	0.995	0.991
Multinomial	100	0.125	0.999	0.999	0.998
Multinomial	100	0.150	1.000	1.000	1.000
Multinomial	100	0.175	1.000	1.000	1.000
Multinomial	100	0.200	1.000	1.000	1.000
Multinomial	100	0.225	1.000	1.000	1.000
Multinomial	100	0.250	1.000	1.000	1.000
Multinomial	100	0.275	1.000	1.000	1.000
Multinomial	100	0.300	1.000	1.000	1.000
Multinomial	100	0.325	1.000	1.000	1.000
Multinomial	100	0.350	1.000	1.000	1.000
Multinomial	100	0.375	1.000	1.000	1.000
Multinomial	100	0.400	1.000	1.000	1.000
Ordinal	25	0.025	0.249	0.385	0.294
Ordinal	25	0.050	0.423	0.547	0.456
Ordinal	25	0.075	0.566	0.647	0.565

Table 4: Simulated power vs. closed-form power (from mean eta-squared/epsilon-squared), by model and sample size. *(continued)*

Model	N	Population Eta2	Power (Actual)	Power (Eta2)	Power (Epsilon2)
Ordinal	25	0.100	0.656	0.759	0.695
Ordinal	25	0.125	0.773	0.839	0.791
Ordinal	25	0.150	0.834	0.882	0.845
Ordinal	25	0.175	0.900	0.915	0.887
Ordinal	25	0.200	0.920	0.950	0.932
Ordinal	25	0.225	0.954	0.967	0.955
Ordinal	25	0.250	0.965	0.979	0.971
Ordinal	25	0.275	0.979	0.985	0.979
Ordinal	25	0.300	0.989	0.991	0.987
Ordinal	25	0.325	0.993	0.994	0.991
Ordinal	25	0.350	0.996	0.996	0.994
Ordinal	25	0.375	0.998	0.997	0.996
Ordinal	25	0.400	1.000	0.998	0.997
Ordinal	50	0.025	0.383	0.490	0.397
Ordinal	50	0.050	0.649	0.733	0.665
Ordinal	50	0.075	0.835	0.873	0.836
Ordinal	50	0.100	0.914	0.936	0.915
Ordinal	50	0.125	0.959	0.973	0.964
Ordinal	50	0.150	0.981	0.987	0.983
Ordinal	50	0.175	0.996	0.995	0.993
Ordinal	50	0.200	0.999	0.998	0.998
Ordinal	50	0.225	0.999	0.999	0.999
Ordinal	50	0.250	0.999	1.000	1.000
Ordinal	50	0.275	1.000	1.000	1.000
Ordinal	50	0.300	1.000	1.000	1.000
Ordinal	50	0.325	1.000	1.000	1.000
Ordinal	50	0.350	1.000	1.000	1.000
Ordinal	50	0.375	1.000	1.000	1.000
Ordinal	50	0.400	1.000	1.000	1.000
Ordinal	75	0.025	0.548	0.650	0.571
Ordinal	75	0.050	0.816	0.858	0.816
Ordinal	75	0.075	0.941	0.959	0.945
Ordinal	75	0.100	0.988	0.989	0.984
Ordinal	75	0.125	0.996	0.997	0.996
Ordinal	75	0.150	0.999	0.999	0.999
Ordinal	75	0.175	1.000	1.000	1.000
Ordinal	75	0.200	1.000	1.000	1.000
Ordinal	75	0.225	1.000	1.000	1.000
Ordinal	75	0.250	1.000	1.000	1.000
Ordinal	75	0.275	1.000	1.000	1.000
Ordinal	75	0.300	1.000	1.000	1.000
Ordinal	75	0.325	1.000	1.000	1.000
Ordinal	75	0.350	1.000	1.000	1.000
Ordinal	75	0.375	1.000	1.000	1.000
Ordinal	75	0.400	1.000	1.000	1.000
Ordinal	100	0.025	0.630	0.720	0.650
Ordinal	100	0.050	0.913	0.943	0.925
Ordinal	100	0.075	0.987	0.988	0.983
Ordinal	100	0.100	0.997	0.998	0.997
Ordinal	100	0.125	1.000	1.000	1.000
Ordinal	100	0.150	1.000	1.000	1.000
Ordinal	100	0.175	1.000	1.000	1.000
Ordinal	100	0.200	1.000	1.000	1.000
Ordinal	100	0.225	1.000	1.000	1.000
Ordinal	100	0.250	1.000	1.000	1.000

Table 4: Simulated power vs. closed-form power (from mean eta-squared/epsilon-squared), by model and sample size. *(continued)*

Model	N	Population Eta2	Power (Actual)	Power (Eta2)	Power (Epsilon2)
Ordinal	100	0.275	1.000	1.000	1.000
Ordinal	100	0.300	1.000	1.000	1.000
Ordinal	100	0.325	1.000	1.000	1.000
Ordinal	100	0.350	1.000	1.000	1.000
Ordinal	100	0.375	1.000	1.000	1.000
Ordinal	100	0.400	1.000	1.000	1.000

Study 2: required sample size (data underlying Figure 5)

Table 5: Required sample size for 80% power, omnibus test: simulated vs. closed-form.

Model	Population R2	Required N (Simulated)	Required N (Closed-form)
Logistic	0.025	312	315
Logistic	0.050	156	157
Logistic	0.075	102	105
Logistic	0.100	77	79
Logistic	0.125	60	63
Logistic	0.150	51	52
Logistic	0.175	42	45
Logistic	0.200	38	39
Multinomial	0.025	246	248
Multinomial	0.050	122	124
Multinomial	0.075	80	83
Multinomial	0.100	59	62
Multinomial	0.125	46	50
Multinomial	0.150	38	41
Multinomial	0.175	32	35
Multinomial	0.200	28	31
Ordinal	0.025	198	198
Ordinal	0.050	98	99
Ordinal	0.075	65	66
Ordinal	0.100	49	50
Ordinal	0.125	39	40
Ordinal	0.150	32	33
Ordinal	0.175	27	28
Ordinal	0.200	24	25

Table 6: Required sample size for 80% power, single-predictor test: simulated vs. closed-form.

Model	Population Eta2	Required N (Simulated)	Required N (Closed-form)
Logistic	0.025	226	226
Logistic	0.050	114	113
Logistic	0.075	75	75
Logistic	0.100	56	57
Logistic	0.125	44	45
Logistic	0.150	37	38
Logistic	0.175	32	32
Logistic	0.200	29	28
Multinomial	0.025	176	175
Multinomial	0.050	87	88

Table 6: Required sample size for 80% power, single-predictor test: simulated vs. closed-form. *(continued)*

Model	Population Eta2	Required N (Simulated)	Required N (Closed-form)
Multinomial	0.075	57	58
Multinomial	0.100	43	44
Multinomial	0.125	34	35
Multinomial	0.150	28	29
Multinomial	0.175	25	25
Multinomial	0.200	23	22
Ordinal	0.025	144	143
Ordinal	0.050	73	71
Ordinal	0.075	48	48
Ordinal	0.100	36	36
Ordinal	0.125	29	29
Ordinal	0.150	25	24
Ordinal	0.175	21	20
Ordinal	0.200	19	18